Statistical properties observed in Japanese Inter-firm Investment network

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In recent years, studies on complex networks with multiple layers are attracting attention [1]. In the study of Japanese Inter-firm network, statistical properties of Trading network between has been analyzed intensively [2], however, little is known about other multiple layers of networks [3]. Here, we construct a double layered network consisting of Japanese Inter-firm Investment network and Trading network, using the data provided by a credit research company, Teikoku Databank.

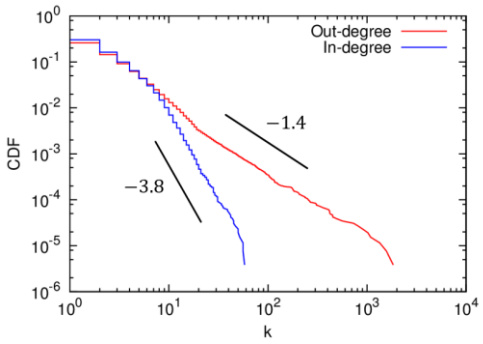


Figure.1. Cumulative distribution of Degrees of Investment network

We confirm that this Investment network has scale-free property with different power exponents for In-degree and Out-degree (Figure.1). We can find another power law relation between the ratio of amount of investment for each firm and the ranking of supplier, in most cases the top 5 firms occupy more than 95 percent of the whole investment (Figure.2). We report on the mutual influence between Inter-firm Investment network and Trading network.

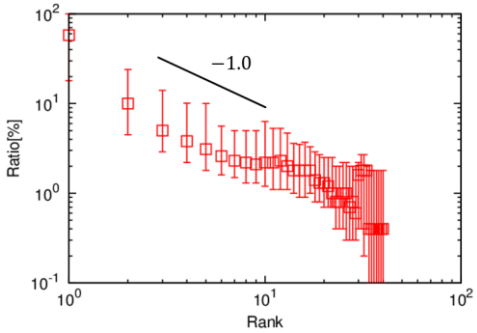


Figure.2. Relation between investment ratio and that rank

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